# PROJECT CONCEPT

## Inspiration and Project Purpose

- Identified a common challenge in learning quantum computing: the heavy reliance on complex mathematics and matrix notation.
- Discovered Bob Coecke's Quantum Picturalism and ZX-calculus, which use visual diagrams to represent quantum logic.
- Developed the idea of creating a tool that makes quantum computing concepts easier to understand through diagrams rather than equations.

## Proposed Solution and Development Approach

- Designed a beginner-friendly Python application that visualises how quantum gates affect qubit states.
- Used Qiskit to model quantum operations such as X, Z, H (Hadamard), and CNOT gates.
- Implemented ZX-calculus-inspired coloured "spiders" and connections to represent quantum logic visually, with an interactive Tkinter interface for animations and user engagement.

## Intended Impact and Applications

- Aims to help students understand quantum concepts such as superposition and entanglement without requiring advanced mathematics.
- Provides a useful educational resource for secondary-school students, university beginners, teachers, STEM clubs, and outreach programmes.
- Demonstrates how visual reasoning and interaction can make complex scientific topics more accessible, engaging, and enjoyable to learn.